Blake Farman

Mathematics and Statistics
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Google Scholar · arXiv · GitHub

Citizenship

United States

Research Interests

Algebra, (Noncommutative) Algebraic Geometry, Derived Categories, Category Theory, and Formalized Mathematics in Lean

Education

Doctor of Philosophy, Mathematics — May 12, 2018
University of South Carolina, Columbia, SC
Dissertation: Geometry of Derived Categories on Noncommutative Projective Schemes
Advisor: Matthew Ballard

Master of Science, Mathematics — May 22, 2011
University of Vermont, Burlington, VT
Major Concentration: Abstract Algebra
Minor Concentration: Algebraic Number Theory

Bachelor of Science, Computer Science — May 16, 2009
Rensselaer Polytechnic Institute, Troy, NY
Minors: Applied Mathematics and French

Appointments

Assistant Professor — September 2024 – Present
Louisiana Tech University, Ruston, LA

Assistant Professor — August 2020 – May 2024
University of Louisiana at Monroe, Monroe, LA

Capital One Endowed Professorship in Mathematics — August 2020 – May 2023
University of Louisiana at Monroe, Monroe, LA

Visiting Assistant Professor — July 2018 – July 2020
Lafayette College, Easton, PA

Graduate Teaching Assistant — August 2012 – May 2018
University of South Carolina, Columbia, SC

Graduate Teaching Assistant — August 2009 – May 2011
University of Vermont, Burlington, VT

Employment

Associate — August 2011 – August 2012
Princeton Consultants, Inc, Princeton, NJ
Information Technology consultant.

Awards and Honors

Education Award 2025
College of Engineering and Science
Louisiana Tech University
With Ann Clifton, for establishing the AXIOM Lab and facilitating the Construction Zone Compound mathematical art installation.

*Project NExT (**N**ew **Ex**periences In **T**eaching) Fellow* Red '22
The Mathematical Association of America

Rensselaer Medalist 2004
Scholarship to Rensselaer Polytechnic Institute ($60,000).

Publications

Clifton, A. W., Farman, B., 2026. WIP: Using Career Panels to Spark Curiosity and Career Awareness Among Mathematics Majors, in: 2026 ASEE Annual Conference & Exposition Proceedings. Presented at the 2026 ASEE Annual Conference & Exposition, ASEE Conferences, Charlotte, North Carolina, p. 60813. https://doi.org/10.18260/1-2--60813.

Clifton, A. W., Fendley, M., Farman, B., Coriell, J., 2025. Work in Progress: First-Year Engineering Students' Confidence in Communicating Mathematical Content, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 57495. https://doi.org/10.18260/1-2--57495.

Farman, B., Clifton, A. W., Long, W. C., Corbett Cruse, K., 2025. Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra, in: 2025 ASEE Annual Conference & Exposition Proceedings. Presented at the 2025 ASEE Annual Conference & Exposition, ASEE Conferences, Montreal, Quebec, Canada, p. 56592. https://doi.org/10.18260/1-2--56592.

Ballard, M., Farman, B., 2021. Kernels for noncommutative projective schemes. J. Noncommut. Geom. 15, 1129–1180. https://doi.org/10.4171/jncg/427.

Farman, B., 2018. Geometry of Derived Categories on Noncommutative Projective Schemes. Ph.D. dissertation. University of South Carolina. https://scholarcommons.sc.edu/cgi/viewcontent.cgi?article=5742&context=etd.

Formalized Mathematics

Peer-reviewed contributions to Lean's `Mathlib` library, developing the basic theory of torsion in abelian categories: Gabriel filters on rings, preradicals and radicals, and torsion theories, together with supporting results on short exact sequences and orthogonal complements.

Introduced torsion theories for abelian categories. October 20, 2026
`CategoryTheory/Abelian/TorsionTheory`; PR #36744

Closure properties of orthogonal complements and S. E. Dickson's theorem. October 8, 2026
`CategoryTheory/ObjectProperty`; PR #44226

Pullbacks and pushouts of short exact sequences. September 30, 2026
`Algebra/Homology/ShortComplex`; PR #44229

Introduced and characterized radicals on abelian categories. March 16, 2026
`CategoryTheory/Abelian/Preradical`; PR #35586

Added Stenström's colon construction for preradicals. March 6, 2026
`CategoryTheory/Abelian/Preradical`; PR #35585

Introduced the basic theory of preradicals on abelian categories. February 26, 2026
`CategoryTheory/Abelian/Preradical`; PR #35514

Defined the topologies associated to ideal filters. February 2, 2026
`RingTheory/IdealFilter`; PR #33853

Introduced ideal filters and Gabriel filters on the lattice of ideals of a ring. January 22, 2026
`RingTheory/IdealFilter`; PR #33021

Generalized `Submodule.colon` to arbitrary subsets and upgraded the surrounding API. January 16, 2026
`RingTheory/Ideal`; PR #33390

Software and Open Educational Resources

dgcat-blueprint 2026 – Present
Lean blueprint and dependency graph for the formalization of dg-categories and the derived category of a dg-category, produced under the Anthropic AI for Science award.
https://github.com/LATechLean/dgcat-blueprint

Funding

Formalizing the dg-Enhancement of Triangulated Categories September 2026
Anthropic AI for Science $20,000
Award of Claude usage credits to formalize dg-categories and the derived category of a dg-category in Lean's `Mathlib` library.
Role: Principal Investigator (PI).

***A**pplied e**X**ploration **I**n **O**pen **M**athematics Lab [AXIOM Lab]* June 2025 – June 2026
Louisiana Board of Regents Support Fund (LEQSF(2025-26)-ENH-DE-07) $60,875
Targeted Departmental Enhancement Grant to establish a dedicated research collaboration lab within the Mathematics and Statistics program at Louisiana Tech University.
Role: Principal Investigator (PI).

Student Activity Enhancement Fee February 2024
Student Development, University of Louisiana at Monroe $2,000
SAEF Grant to take mathematics students to the annual meeting of the Louisiana-Mississippi section of the Mathematical Association of America in New Orleans, LA.

Submitted Grants

Formalizing the Recognition Theorem for Cofibrantly Generated Model Categories in Lean: Foundations for Homotopy Theory and Derived Algebraic Geometry (Declined) October 2025 $69,110
Research Competitiveness Subprogram
Louisiana Board of Regents

LEAPS-MPS: Building the Foundations of Formal (Declined) Homotopy Theory through Model Categories (FY 25) April 2025 $165,753
National Science Foundation

Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined) October 2024 $118,624
DMS - Algebra and Number Theory
National Science Foundation

Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined) November 2023 $113,918
Research Competitiveness Subprogram, FY 24
Louisiana Board of Regents Support Fund

Fourier-Mukai Partners in Noncommutative Algebraic Geometry (Declined) November 2022 $142,041
Research Competitiveness Subprogram, FY 23
Louisiana Board of Regents Support Fund

Derived Categories in Noncommutative Projective Algebraic Geometry (Declined) — $210,000
Proposed NSF Mathematical Sciences Post Doctoral Research Fellowship to work with Ragnar Buchweitz (University of Toronto). Later revised to work with Michael Weems (University of Glasgow)

Dean's Dissertation Fellowship Nominee (Declined) — $25,000
Nominated by the Department of Mathematics at the University of South Carolina for a fellowship from the College of Arts and Sciences to spend a year in residence at the Institute of Advanced Study in Princeton, NJ.

Presentations

Formal Methods in Modern Algebra — October 9, 2025
Computational Analysis and Modeling Seminar
Louisiana Tech University, Ruston, LA

A Little Algebraic Topology — April 11, 2025
Faculty Seminar
COES Graduate Student Council
Louisiana Tech University, Ruston, LA

Fostering Growth Mindsets: Implementing Standards-Based Grading in College Algebra — February 22, 2025
Annual Meeting of the Louisiana/Mississippi Section
Mathematical Association of America
Jackson, MS

Algebra, Geometry, and Categories, Oh My! — May 11, 2023
Louisiana Tech University
Ruston, LA

Implementing Standards Based Grading in STEM — March 28, 2023
University of Louisiana System For our Future Conference 2023
Lake Charles, LA

Algebra, Geometry, and Categories, Oh My! — March 11, 2020
Department Seminar, Lafayette College, Easton, PA

A Little Abstract Nonsense — May 21, 2018
Lafayette College, Easton, PA

Abstract Nonsense: Grokking Category Theory — May 17, 2018
Wilkes University, Wilkes-Barre, PA

Kernels for Noncommutative Projective Schemes — November 28, 2017
Graduate Colloquium
University of South Carolina, Columbia, SC

Kernels for Noncommutative Projective Schemes — October 12, 2017
Algebra Seminar
University of New Brunswick, Fredericton, NB, CA

Kernels for Noncommutative Projective Schemes — October 9, 2017
Algebra Seminar
Temple University, Philadelphia, PA

Kernels for Noncommutative Projective Schemes — August 28, 2017
Algebraic Geometry Seminar
University of South Carolina, Columbia, SC

Noncommutative Projective Schemes — May 13, 2016
Temple Graduate Student Conference in Algebra, Geometry, and Topology
Temple University, Philadelphia, PA

Noncommutative Projective Schemes — May 14, 2016
Student Algebra/Number Theory Seminar
University of South Carolina, Columbia, SC

Fully Faithful Abstract Nonsense: Representable Functors and The Yoneda Embedding — November 2015
Graduate Colloquium
University of South Carolina, Columbia, SC

Fully Faithful Abstract Nonsense: Representable Functors and The Yoneda Embedding — July 2015
Graduate Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar
University of South Carolina, Columbia, SC

Conferences and Workshops

LA/MS Sectional Meeting — February 20 – February 22, 2025
Mathematical Association of America
Jackson, MS

Math Corequisite Academy Workshop — October 15, 2024
Louisiana Board of Regents
Baton Rouge, LA

LA/MS Sectional Meeting — February 29 – March 2, 2024
Mathematical Association of America
New Orleans, LA

High School Math Teacher and Faculty Collaboration — November 30, 2023
Louisiana Department of Education
Baton Rouge, LA

Math Corequisite Academy Workshop — November 7, 2023
Louisiana Board of Regents
Baton Rouge, LA

Mathematical Association of America MathFest — August 1–6, 2023
Tampa, FL

Mathematics Research Community — June 4–10, 2023
Derived Categories, Arithmetic and Geometry, Bridgeland Stability Group
Beaver Hollow Conference Center, Java, NY

Joint Mathematics Meetings — January 3–7, 2023
Boston, MA

Mathematical Association of America MathFest — August 3–6, 2022
Philadelphia, PA

Mathematical Association of America Project NExT (New Experiences in Teaching) — August 1–3, 2022
Philadelphia, PA

Series on Exploring Combinatorics and Number Theory (SECANT) IV — January 15–16, 2022
Cedar Crest College, Allentown, PA (Online)

Derived, Birational, and Categorical Algebraic Geometry — November 14–19, 2021
Invited Participant
Banff International Research Station, Banff, Alberta, CA

Series on Exploring Combinatorics and Number Theory (SECANT) III — January 15–16, 2021
Cedar Crest College, Allentown, PA (Online)

Derived, Birational, and Categorical Algebraic Geometry — November 2–6, 2020
Invited Participant
Banff International Research Station, Banff, Alberta, CA (Online)

Joint Mathematics Meetings — January 15–17, 2020
Denver, CO

Series on Exploring Combinatorics and Number Theory (SECANT) II — October 12–13, 2019
Cedar Crest College, Allentown, PA

Joint Mathematics Meetings — January 16–19, 2019
Baltimore, MD

Series on Exploring Combinatorics and Number Theory (SECANT) — November 17–18, 2018
Cedar Crest College, Allentown, PA

Georgia Algebraic Geometry Symposium — February 23–25, 2018
Georgia Institute of Technology, Atlanta, GA

Workshop on Topics in Algebraic Geometry — November 3–5, 2017
The University of North Carolina at Chapel Hill, Chapel Hill, NC

Palmetto Number Theory Series (PANTS) XXVIII — September 16–17, 2017
University of Tennessee, Knoxville, TN

Georgia Algebraic Geometry Symposium — March 3–5, 2017
University of Georgia, Athens, GA

Workshop on Symplectic Varieties and Geometric Representation Theory — October 28–30, 2016
The University of North Carolina at Chapel Hill, Chapel Hill, NC

UGA-SWAG: Summer Workshop in Algebraic Geometry — August 26–29, 2016
University of Georgia, Athens, GA

Temple Graduate Student Conference in Algebra, Geometry, and Topology — May 14–15, 2016
Temple University, Philadelphia, PA

Georgia Algebraic Geometry Symposium — October 2015
Emory University, Atlanta, GA

AMS Summer Institute on Algebraic Geometry — July 27–31, 2015
University of Utah, Salt Lake City, UT

Georgia Algebraic Geometry Symposium — October 2014
University of Georgia, Athens, GA

Arizona Winter School — March 2014
University of Arizona, Tucson, AZ

Palmetto Number Theory Series (PANTS) XXI — December 2013
Clemson University, Clemson, SC

Palmetto Number Theory Series (PANTS) XX — September 2013
Davidson College, Davidson, NC

Algebraic Geometry Northeastern Series (AGNES) — April 2013
Yale University, New Haven, CT

AMS Spring Western Sectional Meeting — April 2013
University of Colorado, Boulder, CO

Palmetto Number Theory Series (PANTS) XIX — December 2012
University of South Carolina, Columbia, SC

Palmetto Number Theory Series (PANTS) XVIII — September 2012
Wake Forest University, Winston-Salem, NC

Graduate Mentoring and Research Supervision

Emma Foster 2026 – Present
Ph.D. Student (Committee Chair), Computational Analysis and Modeling
Louisiana Tech University

Joshua Bush 2025–2026
Ph.D. Student (Committee Chair), Computational Analysis and Modeling
Louisiana Tech University

Josiah Norman Spring 2026
M.S. Student (Committee Member)
Louisiana Tech University

Undergraduate Mentoring and Research Supervision

Joshua Robinson Spring 2025
Louisiana Tech University
Co-mentored with Ann Clifton on a Cayley graph research project, *Structural and Topological Properties of Cayley Graphs*, presented at the Louisiana Tech Undergraduate Research & Service-Learning Symposium and awarded second place in the Student Paper Competition at the LA/MS Section meeting of the Mathematical Association of America.

Riya Shrestha Spring 2024
Emerging Scholars Program, University of Louisiana at Monroe
Knot Theory research project focused on identifying a family of knots for which there is a winning strategy in the Region Unknot Game.

The Entrepreneurship Pelican Cup Spring 2023
Immerse AR
Served as faculty advisor for one of six finalists in the Pelican Cup Entrepreneurship business competition.

Sagar Bhetuwal Spring 2021
Emerging Scholars Program, University of Louisiana at Monroe
Number Theory research project focused on providing a formal verification of the proof of the Prime Number Theorem using the Lean proof assistant.

Organizer

Student Algebra/Number Theory Seminar Summer 2016
University of South Carolina

Student Algebraic Geometry, Commutative Algebra, and Number Theory Seminar Summer 2015
University of South Carolina

Student Algebraic Geometry Seminar Spring 2015
University of South Carolina

Student Algebraic Geometry Seminar Fall 2014
University of South Carolina

Teaching Innovations

Interactive Theorem Proving with Lean, Math 490/587 Spring 2025
Designed and taught a new course introducing undergraduate and graduate students to formal mathematics in the Lean proof assistant. Course materials released publicly at https://github.com/LATechLean/ITPLxS25.

Construction Zone Compound, Mathematical Art Installation — Spring 2025
With Ann Clifton and Glen Whitney (Studio Infinity), guided students in building a compound of five tetrahedra from road-cone barriers and radio-antenna mounting hardware, exhibited alongside the senior design showcase. Recognized, together with the AXIOM Lab, by the 2025 COES Education Award.

Teaching Experience

Parenthesized numbers give the number of sections taught that term.

Louisiana Tech University

Course Number	Course Name	Terms
Math 587	Interactive Theorem Proving with Lean	S25
Math 5843	Algebraic Geometry	S26
Math 5083	Intro to Abstract Algebra	F25, F26
Math 4903	Algebraic Geometry	S26
Math 490	Interactive Theorem Proving with Lean	S25
Math 4163	Abstract Algebra	W27
Math 4083	Intro to Abstract Algebra	F25, F26
Math 308	Introduction to Linear Algebra	W25
Math 243	Calculus III	F24

University of Louisiana at Monroe

Course Number	Course Name	Terms
Math 4007	History of Mathematics	F22
Math 3086	Modern Algebra	F21, S23
Math 3080	Real Analysis	S22, F23
Math 3073	Numerical Methods	F23
Math 3040	Introduction to Algebra and Analysis	S24
Math 2040	Foundations of Mathematics	S23
Math 2032	Calculus III	F22
Math 2002	Applied Linear Algebra	S21, F21, S23
Math 1031	Calculus I	S24
Math 1018	Contemporary Mathematics	F21 (2), S22 (2)
Math 1011	College Algebra	F20 (3), S21 (2), S24 (2)
Math 1007	Contemporary Mathematics	F22 (2), S23, F23 (2)

Lafayette College

Course Number	Course Name	Terms
Math 272	Linear Algebra with Applications	S20
Math 162	Calculus II	S19 (2), F19 (2)
Math 161	Calculus I	F18 (2), F19, S20 (2)
Math 161	Summer Program to Advance Leadership	Su19

University of South Carolina

Course Number	Course Name	Terms
Math 170	Finite Mathematics	S14, S16

Course Number	Course Name	Terms
Math 142	Calculus II	Su14, S18
Math 122	Calculus for Business Administration and Social Sciences	S17
Math 116	Brief Precalculus Mathematics	F13 (2), F15 (2), F16 (2)
Math 115	Precalculus Mathematics	F14
Math 111	Basic College Mathematics	Su13, S15, F17

University of Vermont

Course Number	Course Name	Terms
Math 019	Fundamentals of Calculus I	F10 (2)
Math 017	Applications of Finite Math	F09, S10, Su10

Administrative Appointments

Coordinator, M.S. in Mathematics — September 2026 – Present
Mathematics and Statistics
Louisiana Tech University

Leadership and Governance

Communication Coordinator (Officer) — February 2025 – Present
Louisiana-Mississippi Section
Mathematical Association of America

Membership Outreach Director (Officer) — January 2025 – Present
Board of Directors
Friends of Black Bayou
Black Bayou National Wildlife Refuge
Monroe, LA

Director at Large — January 2024 – January 2025
Board of Directors
Friends of Black Bayou
Black Bayou National Wildlife Refuge
Monroe, LA

Service

To the Profession

Section NExT Committee Member — February 2025 – Present
Louisiana-Mississippi Section
Mathematical Association of America

To the State

Mindset Meauxtivator — Fall 2024 – Spring 2025
Meauxmentum Initiative, Louisiana Board of Regents, with Motivate Lab (University of Virginia)
One of thirteen Louisiana faculty trained in growth-mindset pedagogy for corequisite courses.

To the University

Louisiana Tech University

Faculty Senate — 2025–2026
College of Engineering and Science Representative; member of the subcommittee revising Policy 1301

Undergraduate Advising — 2025–2026
Advisees: J. D. Vessel and Bryce Labbie

Career Panels for Mathematics Majors — 2025–2026
Collaborated with Ann Clifton and the Mathematics Club to organize career panels for mathematics majors.

Literary Rally — March 15, 2025
Proctor

Undergraduate Research & Service-Learning Symposium — March 14, 2025
Judge

Professors on the Plaza — October 16, 2024

University of Louisiana at Monroe

Mathematics Hiring Committee Chair — Fall 2023

Mathematics Hiring Committee Member — Fall 2022 – Spring 2023

Browse on the Bayou — September 30, 2022
Mathematics program representative for ULM's Fall on-campus recruiting event

Co-Requisite Course Design — Spring 2022
Contemporary Mathematics

Browse on the Bayou — March 21, 2022
Mathematics program representative for ULM's Spring on-campus recruiting event

Investiture Ceremony Marshal — October 1, 2021

University Curriculum Committee Member — 2021–2024

Louisiana Math Forward Committee Member — 2021

Mathematics Hiring Committee Member — Spring 2021

University of South Carolina

High School Math Contest Proctor — 2013–2015

To the Community

Fall Celebration — October 17, 2026
Represented Friends of Black Bayou as Membership Outreach Director.
Black Bayou Lake National Wildlife Refuge

National Hunting and Fishing Day — September 26, 2026
Represented Friends of Black Bayou at the Louisiana Department of Wildlife and Fisheries event for National Hunting and Fishing day.
Black Bayou Lake National Wildlife Refuge

National Hunting and Fishing Day — September 27, 2025
Represented Friends of Black Bayou at the Louisiana Department of Wildlife and Fisheries event for National Hunting and Fishing day.
Black Bayou Lake National Wildlife Refuge

Fall Celebration October 19, 2024
Setup, Operated Canoe/Kayak Rental, Breakdown
Black Bayou Lake National Wildlife Refuge

Water Sweep September 14, 2024
Waterway cleanup at Black Bayou Lake National Wildlife Refuge
Ouachita Green

Professional Membership

Mathematical Association of America Member

American Mathematical Society Member